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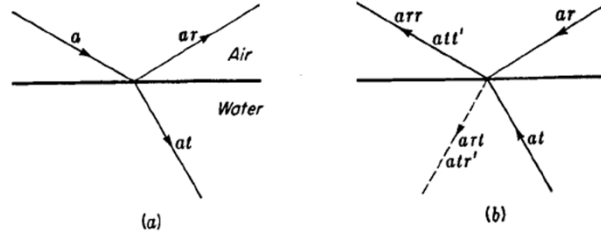
INTERFERENCE INVOLVING
MULTIPLE REFLECTIONS

FIGURE 14A
Stokes' treatment of reflection.

Thus, let the symbol a in the left-hand part of Fig. 14A represent the amplitude (not the intensity) of a set of waves striking the surface, let r be the fraction of the amplitude reflected, and let t be the fraction transmitted. The amplitudes of the two sets of waves will then be ar and at , as shown. Now, following a treatment given by Stokes*, imagine the two sets reversed in direction, as in part (b) of the figure. Provided there is no dissipation of energy by absorption, a wave motion is a strictly reversible phenomenon. It must conform to the law of mechanics known as the *principle of reversibility*, according to which the result of an instantaneous reversal of all the velocities in a dynamic system is to cause the system to retrace its whole previous motion. That the paths of light rays are in conformity with this principle has already been stated in Sec. 1.8. The two reversed trains, of amplitude ar and at , should accordingly have as their net effect after striking the surface a wave in air equal in amplitude to the incident wave in part (a) but traveling in the opposite direction. The wave of amplitude ar gives a reflected wave of amplitude arr and a refracted wave of amplitude art . If we call r' and t' the fractions of the amplitude reflected and refracted when the reversed wave at strikes the boundary from below, this contributes amplitudes att' and atr' to the two waves, as indicated. Now, since the resultant effect must consist only of a wave in air of amplitude a , we have

$$att' + arr = a \quad (14a)$$

$$art + atr' = 0 \quad (14b)$$

The second equation states that the two incident waves shall produce no net disturbance on the water side of the boundary. From Eq. (14a) we obtain

$$tt' = 1 - r^2 \quad (14c)$$

and from Eq. (14b)

$$r' = -r \quad (14d)$$

It might at first appear that Eq. (14c) could be carried further by using the fact that intensities are proportional to squares of amplitudes and by writing, by conservation of energy, $r^2 + t^2 = 1$. This would immediately yield $t = t'$. The result is not correct, however, for two reasons: (1) although the proportionality of intensity with square of amplitude holds for light traveling in a single medium, passage into a different medium brings in the additional factor of the index of refraction in determining the intensity; (2) it is not to the intensities that the conservation law is to be applied but to the total energies of the beams. When there is a change in width of the beam, as in refraction, it must also be taken into account.

The second of Stokes' relations, Eq. (14d), shows that the *reflectance*, or fraction of the intensity reflected, is the same for a wave incident from either side of the boundary, since the negative sign disappears upon squaring the amplitudes. It should be noted, however, that the waves must be incident at angles such that they correspond to angles of incidence and refraction. The difference in sign of the amplitudes in Eq. (14d) indicates a difference of phase of π between the two cases, since a reversal of sign means a displacement in the opposite sense. If there is no phase change on reflection from above, there must be a phase change of π on reflection from below; or correspondingly, if there is no change on reflection from below, there must be a change of π on reflection from above.

The conclusion reached above about the change of phase is not dependent on the applicability of the principle, i.e., on the absence of absorption, but holds for reflection from any boundary. It is a matter of experimental observation that in the reflection of light under the above conditions, the phase change of π occurs when the light strikes the boundary from the side of higher velocity,* so that the second of the two alternatives mentioned is the correct one in this case. A change of phase of the same type is encountered in the reflection of simple mechanical waves, such as transverse waves in a rope.

14.1 REFLECTION FROM A PLANE-PARALLEL FILM

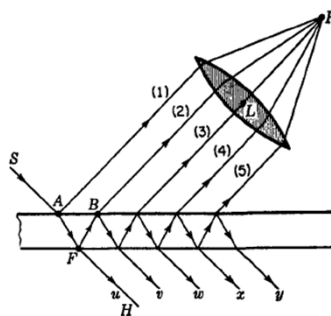


FIGURE 14B
Multiple reflections in a plane-parallel film.

Let a ray of light from a source S be incident on the surface of such a film at A (Fig. 14B). Part of this will be reflected as ray 1 and part refracted in the direction AF . Upon arrival at F , part of the latter will be reflected to B and part refracted toward H . At B the ray FB will be again divided. A continuation of this process yields two sets of parallel rays, one on each side of the film. In each of these sets, of course, the intensity decreases rapidly from one ray to the next. If the set of parallel reflected rays is now collected by a lens and focused at the point P , each ray will have traveled a different distance, and the phase relations may be such as to produce destructive or constructive interference at that point. It is such interference that produces the colors of thin films when they are viewed by the naked eye. In such a case L is the lens of the eye, and P lies on the retina.

In order to find the phase difference between these rays, we must first evaluate the difference in the optical path traversed by a pair of successive rays, such as rays 1 and 2. In Fig. 14C let d be the thickness of the film, n its index of refraction, λ the wavelength of the light, and ϕ and ϕ' the angles of incidence and refraction. If BD is perpendicular to ray 1, the optical paths from D and B to the focus of the lens will be equal. Starting at A , ray 2 has the path AFB in the film and ray 1 the path AD in air. The difference in these optical paths is given by

$$\Delta = n(ABF) - AD$$

If BF is extended to intersect the perpendicular line AE at G , $AF = GF$ because of the equality of the angles of incidence and reflection at the lower surface. Thus we have

$$\Delta = n(GB) - AD = n(GC + CB) - AD$$

Now AC is drawn perpendicular to FB ; so the broken lines AC and DB represent two successive positions of a wave front reflected from the lower surface. The optical paths must be the same by any ray drawn between two wave fronts; so we may write

$$n(CB) = AD$$

The path difference then reduces to

$$\Delta = n(GC) = n(2d \cos \phi') \quad (14e)$$

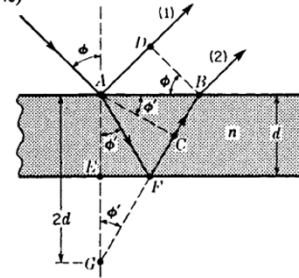


FIGURE 14C
Optical-path difference between two consecutive rays in multiple reflection (see Fig. 14A).

If this path difference is a whole number of wavelengths, we might expect rays 1 and 2 to arrive at the focus of the lens in phase with each other and produce a maximum of intensity. However, we must take account of the fact that ray 1 undergoes a phase change of π at reflection, while ray 2 does not, since it is internally reflected. The condition

$$\bullet \quad 2nd \cos \phi' = m\lambda \quad \text{Minima} \quad (14f)$$

then becomes a condition for *destructive* interference as far as rays 1 and 2 are concerned. As before, $m = 0, 1, 2, \dots$ is the order of interference.

Next we examine the phases of the remaining rays, 3, 4, 5, Since the geometry is the same, the path difference between rays 3 and 2 will also be given by Eq. (14e), but here there are only internal reflections involved, so that if Eq. (14f) is fulfilled, ray 3 will be in the same phase as ray 2. The same holds for all succeeding pairs, and so we conclude that under these conditions rays 1 and 2 will be out of phase, but rays 2, 3, 4, will be in phase with each other. On the other hand, if conditions are such that

$$\bullet \quad 2nd \cos \phi' = (m + \frac{1}{2})\lambda \quad \text{Maxima} \quad (14g)$$

ray 2 will be in phase with 1, but 3, 5, 7, ... will be out of phase with 2, 4, 6, Since 2 is more intense than 3, 4 more intense than 5, etc., these pairs cannot cancel each other, and since the stronger series combines with 1, the strongest of all, there will be a maximum of intensity.

For the minima of intensity,

For the minima of intensity, ray 2 is out of phase with ray 1, but 1 has a considerably greater amplitude than 2, so that these two will not completely annul each other. We can now prove that the addition of 3, 4, 5, . . . , which are all in phase with 2, will give a net amplitude just sufficient to make up the difference and to produce complete darkness at the minima. Using a for the amplitude of the incident wave, r for the fraction of this reflected, and t or t' for the fraction transmitted in going from rare to dense or dense to rare, as was done in Stokes' treatment of reflection, Fig. 14D is constructed and the amplitudes labeled as shown. In accordance with Eq. (14d), we have taken the fraction reflected internally and externally to be the same. Adding the amplitudes of all the reflected rays but the first on the upper side of the film, we obtain the resultant amplitude,

$$\begin{aligned} A &= atrt' + atr^3t' + atr^5t' + atr^7t' + \cdots \\ &= atrt' (1 + r^2 + r^4 + r^6 + \cdots) \end{aligned}$$

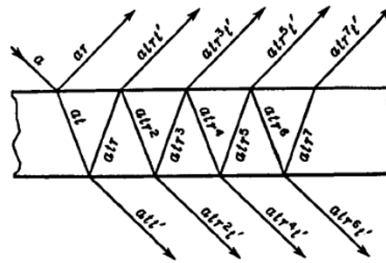


FIGURE 14D
Amplitudes of successive rays in multiple reflection.

Since r is necessarily less than 1, the geometrical series in parentheses has a finite sum equal to $1/(1 - r^2)$, giving

$$A = atrt' \frac{1}{1 - r^2}$$

But from Stokes' treatment, Eq. (14c), $tt' = 1 - r^2$; so we obtain finally

$$A = ar \quad (14h)$$

This is just equal to the amplitude of the first reflected ray, so we conclude that under the conditions of Eq. (14f) there will be complete destructive interference.

14.2 FRINGES OF EQUAL INCLINATION

If the image of an *extended* source reflected in a thin plane-parallel film is examined, it will be found to be crossed by a system of distinct interference fringes, provided the source emits monochromatic light and provided the film is sufficiently thin. Each bright fringe corresponds to a particular path difference giving an integral value of m in Eq. (14g). For any fringe, the value of ϕ is fixed; so the fringe will have the form of the arc of a circle whose center is at the foot of the perpendicular drawn from the eye to the plane of the film. Evidently we are here concerned with fringes of equal inclination, and the equation for the path difference has the same form as for the circular fringes in the Michelson interferometer (Sec. 13.9).

Note that if m is the order of interference for light incident on the film at $\phi = 0^\circ$, Eq. (14f) gives

$$m = \frac{2nd}{\lambda}$$

which would be a dark fringe. Since the path difference for the first, second, and third, etc., bright fringes will be at progressively larger angles of ϕ and ϕ' [Eq. (14g)], the successive path differences, $2nd \cos \phi'$, will be successively shorter and bright-light fringes will be at angles where $2nd \cos \phi'$ is equal $(m - \frac{1}{2})\lambda$, $(m - \frac{3}{2})\lambda$, $(m - \frac{5}{2})\lambda$, etc.

The necessity of using an extended source will become clear upon consideration of Fig. 14B. If a very distant point source S is used, the parallel rays will necessarily reach the eye at only one angle (that required by the law of reflection) and will be focused to a point P . Thus only one point will be seen, either bright or dark, according to the phase difference at this particular angle.

These fringes are seen by the eye only if the film is very thin, unless the light is reflected practically normal to the film. At other angles, since the pupil of the eye has a small aperture, increasing the thickness of the film will cause the reflected rays to get so far apart that only one enters the eye at a time. Obviously no interference can occur under these conditions. Using a telescope of large aperture, the lens may include enough rays for the fringes to be visible with thick plates, but unless viewed nearly normal to the plate, they will be so finely spaced as to be invisible. The fringes seen with thick plates near normal incidence are often called *Haidinger* fringes*.

* W. K. von Haidinger (1795–1871). Austrian mineralogist and geologist, for 17 years director of the Imperial Geological Institute in Vienna.

14.3 INTERFERENCE IN THE TRANSMITTED LIGHT

The rays emerging from the lower side of the film, shown in Fig. 14B and 14D, can also be brought together with a lens and caused to interfere. Here, however, there are no phase changes at reflection for any of the rays, and the relations are such that Eq. (14f) now becomes the condition for maxima and Eq. (14g) the condition for minima. For maxima, the rays $u, v, w, \dots$ of Fig. 14B are all in phase, while for minima, $v, x, \dots$ are out of phase with $u, w, \dots$. When the reflectance r^2 has a low value, as with the surfaces of unsilvered glass, the amplitude of u is much the greatest in the series and the minima are not by any means black. Figure 14E shows quantitative curves for the intensity transmitted I_T and reflected I_R plotted in this instance for $r = 0.2$ according to Eqs. (14n) and (14o), ahead. The corresponding reflectance of 4 percent is close to that of glass at normal incidence. The abscissas δ in the figure represent the phase difference between successive rays in the transmitted set or between all but the first pair in the reflected set, which by Eq. (14e) is

$$\delta = k\Delta = \frac{2\pi}{\lambda} \Delta = \frac{4\pi}{\lambda} nd \cos \phi' \quad (14i)$$

It will be noted that the curve for I_R looks very much like the $\cos^2$ contour obtained from the interference of two beams. It is not exactly the same, however, and the resemblance holds only when the reflectance is small. Then the first two reflected beams are so much stronger than the rest that the latter have little effect. The important changes that come in at higher values of the reflectance will be discussed in Sec. 14.7.

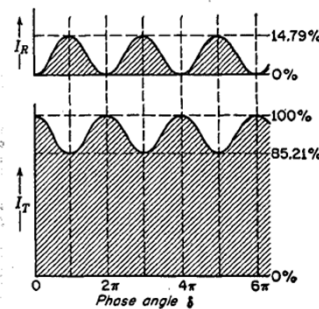


FIGURE 14E
Intensity contours of the reflected and transmitted fringes from a film having a reflectance of 4 percent.

14.4 FRINGES OF EQUAL THICKNESS

If the film is not plane-parallel, so that the surfaces make an appreciable angle with each other as in Fig. 14F(a), the interfering rays do not enter the eye parallel to each other but appear to diverge from a point near the film. The resulting fringes resemble the localized fringes in the Michelson interferometer and appear to be formed in the film itself. If the two surfaces are plane, so that the film is wedge-shaped, the fringes will be practically straight, following the lines of equal thickness. In this case the path difference for a given pair of rays is practically that given by Eq. (14e). Provided that observations are made almost normal to the film, the factor $\cos \phi'$ may be considered equal to 1, and the condition for bright fringes becomes

$$2nd = (m + \frac{1}{2})\lambda \quad (14j)$$

In going from one fringe to the next m increases by 1, and this requires the optical thickness of the film nd to change by $\lambda/2$.

Fringes formed in thin films are easily shown in the laboratory or lecture room by using two pieces of ordinary plate glass. If they are laid together with a thin strip of paper along one edge, we obtain a wedge-shaped film of air between the plates. When a sodium flame or arc is viewed as in Fig. 14F, yellow fringes are clearly seen. If a carbon arc and filter are used, the fringes may be projected on a screen with a lens. On viewing the reflected image of a monochromatic source, one will find it to be crossed by more or less straight fringes, such as those in Fig. 14F(b).

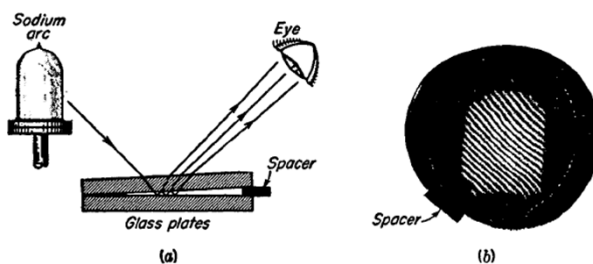


FIGURE 14F
Fringes of equal thickness: (a) method of visual observation; (b) photograph taken with a camera focused on the plates.

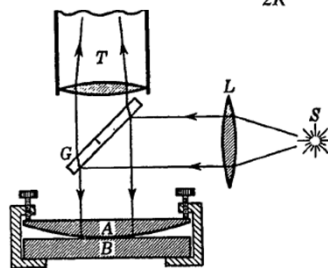
14.5 NEWTON'S RINGS

If the fringes of equal thickness are produced in the air film between a convex surface of a long-focus lens and a plane glass surface, the contour lines will be circular. The ring-shaped fringes thus produced were studied in detail by Newton,* although he was not able to explain them correctly. For purposes of measurement, the observations are usually made at normal incidence by an arrangement such as that in Fig. 14G, where the glass plate G reflects the light down on the plates. After reflection, it is transmitted by G and observed in the low-power microscope T . Under these conditions the positions of the maxima are given by Eq. (14j), where d is the thickness of the air film. Now if we designate by R the radius of curvature of the surface A and assume that A and B are just touching at the center, the value of d for any ring of radius r is the sagitta of the arc, given by

$$d = \frac{r^2}{2R} \quad (14k)$$

FIGURE 14G

Experimental arrangement used in viewing and measuring Newton's rings.



Substitution of this value in Eq. (14j) will then give a relation between the radii of the rings and the wavelength of the light. For quantitative work, one may not assume the plates to barely touch at the point of contact, since there will always be either some dust particles or distortion by pressure. Such disturbances will merely add a small constant to Eq. (14k), however, and their effect can be eliminated by measuring the diameters of at least two rings.

Because the ring diameters depend on wavelength, white light will produce only a few colored rings near the point of contact. With monochromatic light, however, an extensive fringe system such as that shown in Fig. 14H is observed. When the contact is perfect, the central spot is black. This is direct evidence of the relative phase change of π between the two types of reflection, air-to-glass and glass-to-air, mentioned in Sec. 14.1. If there were no such phase change, the rays reflected from the two surfaces in contact should be in the same phase and produce a bright spot at the center.

In an interesting modification of the experiment, due to Thomas Young, the lower plate has a higher index of refraction than the lens, and the film between is filled with an oil of intermediate index. Then both reflections are at "rare-to-dense" surfaces, no relative phase change occurs, and the central fringe of the reflected system is bright. The experiment does not tell us at which surface the phase change in the ordinary arrangement occurs, but it is now definitely known (Sec. 25.4) that it occurs at the lower (air-to-glass) surface.

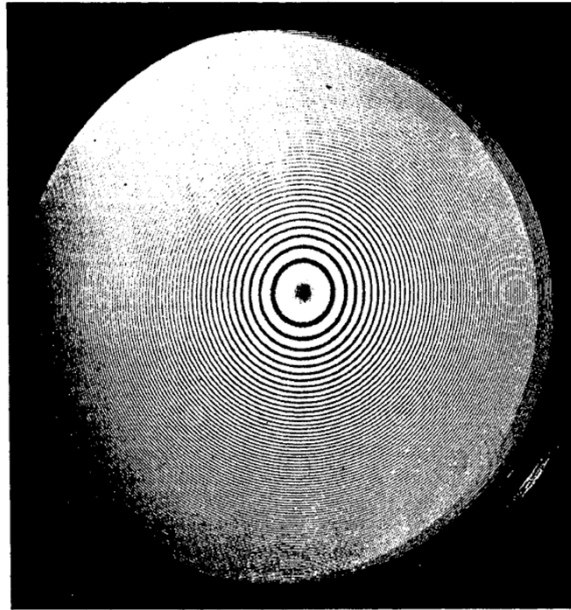


FIGURE 14H
Newton's rings. (By permission from Bausch & Lomb Incorporated.)

application of the principles of interference

14.6 NONREFLECTING FILMS

A simple and very important application of the principles of interference in thin films has been the production of *coated* surfaces. If a film of a transparent substance of refractive index n' is deposited on glass of a larger index n to a thickness of one-quarter of the wavelength of light in the film, so that

$$d = \frac{\lambda}{4n'}$$

the light reflected at normal incidence is almost completely suppressed by interference. This corresponds to the condition $m = 0$ in Eq. (14g), which here becomes a condition for *minima* because the reflections at both surfaces are "rare-to-dense." The waves reflected from the lower surface have an extra path of one-half wavelength over those from the upper surface, and the two, combined with the weaker waves from multiple reflections, therefore interfere destructively. For the destruction to be complete, however, the fraction of the amplitude reflected at each of the two surfaces must be exactly the same, since this specification is made in proving the relation of Eq. (14h). It will be true for a film in contact with a medium of higher index only if the index of the film obeys the relation

$$n' = \sqrt{n}$$

multilayers, may also be used, with suitable thickness, to accomplish the opposite purpose, namely to increase the reflectance. They may be used, for example, as beam-splitting mirrors to divide a beam of light into two parts of a given intensity ratio.

14.7 SHARPNESS OF THE FRINGES

As the reflectance of the surfaces is increased, either by the above method or by lightly silvering them, the fringes due to multiple reflections become much narrower. The striking changes that occur are shown in Fig. 14I, which is plotted for $r^2 = 0.04$, 0.50, and 0.80 according to the theoretical equations to be derived below. The curve labeled 4% is just that for unsilvered glass which was given in Fig. 14E. Since, in the absence of any absorption, the intensity transmitted must be just the complement of that reflected, the same plot will represent the contour of either set. One is obtained from the other by merely turning the figure upside down or by inverting the scale of ordinates, as shown by the down arrow at the right in Fig. 14I.

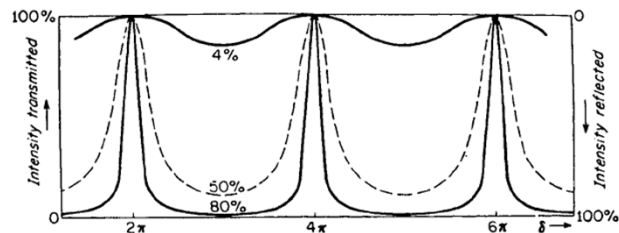


FIGURE 14I
Intensity contours of fringes due to multiple reflections, showing how the sharpness depends on reflectance.